

Rocksdb_dataset-9

Commit 161:

48eb646787437bd9a1c5f03b717038ce23e15ee5

Non-C file modifications, therefore no performance regression testing is required.

Commit 162:

be99011f08a042ad76859476b2263ce108ec3be8

Non-C file modifications, therefore no performance regression testing is required.

Commit 163:

b7a9d414c8bbcc7a5dd9329e344a0451770ab555

The 1st symbolic execution

Testcase:for (int i = 0; i < 10000; i++){key = "key_exit" + std::to_string(i + 30001);value = "new_value" + key;db->Put(write_options_sync, key, value);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  323453.88802127657
4  The new version is executed by default
5  316232.51566666666
6  The new version
7  320212.6675625
8  Comparison of New and old versions (along symbolic execution path)
9  0.010122086935064627
10 Comparison of New and old versions (along the default execution path)
11 -0.012429714058880945
```

No performance regression

The 2nd symbolic execution

Testcase:for (int i = 0; i < 10000; i++){key = "key_exit" + std::to_string(i + 30001);value = "new_value" + key;db->Put(write_options_sync, key, value);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  317197.47104255325
4  The new version is executed by default
5  316559.65502127656
6  The new version
7  308963.8468125
8  Comparison of New and old versions (along symbolic execution path)
9  0.02664915107381477
10 Comparison of New and old versions (along the default execution path)
11 0.0245847800224543
12
13
14 icount diff
15 Absolute Difference: 7160.00
16 Percentage Difference: 78.54%
```

There may be performance regression

Commit 164:

30e097e365bd00d15144f1e06632a8c8e4f54512

Non-C file modifications, therefore no performance regression testing is required.

Commit 165:

5735ff4e0322fe82e1201b4df18afae445365c42

Non-C file modifications, therefore no performance regression testing is required.

Commit 166:

24e2b05e6179fda5770f3a4203fa7cdfda7fd936

The 1st symbolic execution

Testcase:for(int i = 0; i< 1000000; i++){key = "key_exit" + std::to_string(i);db->Get(rocksdb::ReadOptions(), key, &value);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1266716.2602857144
4  The new version is executed by default
5  1250056.376387755
6  The new version
7  1259266.1235957446
8  Comparison of New and old versions (along symbolic execution path)
9  0.005916252768474761
10 Comparison of New and old versions (along the default execution path)
11 -0.00731358291581124
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i=0; i < 50; i++){rocksdb::Iterator* it = db->NewIterator(rocksdb::ReadOptions());for (it->SeekToLast(); it->Valid(); it->Prev()) {} delete it;}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  3327916.1364081637
4  The new version is executed by default
5  3351179.6303695654
6  The new version
7  3406340.7622083337
8  Comparison of New and old versions (along symbolic execution path)
9  -0.023023129884787937
10 Comparison of New and old versions (along the default execution path)
11 -0.016193662257973054
```

No performance regression

The 3rd symbolic execution

Testcase:for(int i =0;i < 100; i++){rocksdb::Iterator* it = db->NewIterator(rocksdb::ReadOptions());for (it->SeekToFirst(); it->Valid(); it->Next()) {} delete it;}

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  4441153.176714285
4  The new version is executed by default
5  4427776.48802
6  The new version
7  4532549.26288
8  Comparison of New and old versions (along symbolic execution path)
9  -0.020164388926606297
10 Comparison of New and old versions (along the default execution path)
11 -0.023115639518372755
```

No performance regression

The 4th symbolic execution

Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"), end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db->DefaultColumnFamily(), start_key, end_key);rocksdb::Slice start_key1("key_exit:200");rocksdb::Slice end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db->CompactRange(comp_options, &start_key1, &end_key2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1721469.6717083333
4  The new version is executed by default
5  1706017.8347291667
6  The new version
```

7	1704029.8690416666
8	Comparison of New and old versions (along symbolic execution path)
9	0.010234446580724959
10	Comparison of New and old versions (along the default execution path)
11	0.0011666260806907668

No performance regression

Commit 167:
4069afeede6363a2f723ea72a6b24bd10fe0d0c7

Unable to provide test cases